

Benchmarking Neural Surrogates for Hydrofoil RANS Flow, Cavitation-Risk Prediction, and Hydrodynamic Shape Optimization

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Abstract—Neural flow surrogates are usually compared by field error, although their practical value often lies in the designs they select. We compare a convolutional neural network (CNN) U-Net, a physics-regularized point network, a Fourier neural operator (FNO), and a deep operator network (DeepONet) on 381 OpenFOAM simulations of hydrofoils in water. Sixty-four cases from two unseen NACA families form the validation set. The models predict Reynolds-averaged Navier-Stokes (RANS) fields, lift, drag, and pressure-based cavitation-inception risk before driving the same four-parameter shape search. The physics-regularized network gives the best pressure accuracy ($R^2 = 0.966$), cavitation F1 score (0.921), and inference speed; DeepONet gives the most accurate force coefficients. Inference is 917–1801 times faster than computational fluid dynamics (CFD). Fresh OpenFOAM simulations confirm lift-to-drag ratio (L/D) gains of 2.3–79.7% for the four selected designs, while a matched direct-CFD search reaches $L/D = 29.89$. Three surrogates select CFD-verified designs above that local-search reference, with the best reaching 45.96. Field accuracy, force accuracy, and optimization performance are related but not interchangeable.

Index Terms—artificial intelligence for science, CFD surrogate, neural operator, physics-informed learning, hydrofoil optimization, cavitation inception

I. Introduction

For a hydrofoil operating in water, shape design is an optimization problem wrapped around a flow solver. Each change in camber, thickness, or incidence requires another Reynolds-averaged Navier-Stokes (RANS) solution, so even a modest search can consume hundreds of computational fluid dynamics (CFD) evaluations; neural surrogates move most of that search off the solver by learning the map from geometry and operating conditions to flow and forces [1]. The difficult question is not simply which network minimizes field error, but which one preserves the quantities that determine a useful design.

The convolutional neural network U-Net (CNN-U-Net), Fourier neural operator (FNO), deep operator network (DeepONet), and physics-regularized point network impose very different structure on that map [2], [3], [4], [5]. A convolutional model emphasizes local multiscale features; FNO mixes information globally in Fourier space; DeepONet separates parameter and coordinate representations; and a physics penalty constrains a pointwise regressor.

Their relative merits may therefore change as the target moves from pressure fields to forces and finally to shape optimization.

Airfoil-surrogate research has progressed from field prediction [6], [7], [8], [9] to benchmarks that evaluate surface forces, out-of-distribution accuracy, and computational cost [10], [11], [12]. Those metrics establish whether a model approximates CFD, but they do not by themselves establish whether it will make the same design decision. A small, structured force error can be amplified when an optimizer repeatedly queries a narrow region of design space. The missing experiment is a controlled comparison that follows several surrogate families through the same optimizer and then checks the selected geometries with new CFD solutions. High-fidelity endpoint verification is standard in surrogate shape optimization, while a matched direct-CFD search provides a stronger reference when its cost is manageable [13], [14].

That experiment is the contribution of this paper; we do not propose a new architecture. We compare four representative families on the same RANS data, complete-geometry validation split, targets, and three-start hydrodynamic shape optimization (HSO) problem. The benchmark connects field and force accuracy, pressure-threshold cavitation screening, and runtime to the final geometry selected by each model. Fresh OpenFOAM solutions measure endpoint discrepancy, and a direct OpenFOAM search uses the same starts and optimization budget. Together, these tests ask whether validation accuracy predicts search quality and how closely each inexpensive search approaches the CFD-driven result.

II. Related Work

Convolutional surrogates established that steady flow fields can be learned directly from geometry encodings on Cartesian grids [6], [7], [9]. Later datasets and benchmarks broadened evaluation beyond image error: AIRFRANS includes surface quantities and out-of-distribution splits, the geometric neural operator (GINO) addresses irregular three-dimensional geometries, and ML4CFD evaluates modern operator-learning systems under a common pro-

tocol [10], [11], [12]. These studies motivate our geometry-level validation split and separate reporting of fields, forces, runtime, and low-pressure classification.

Neural operators represent an entire parameter-to-solution map rather than a single flow realization. FNO performs global spectral mixing on a regular grid, whereas DeepONet combines a parameter-dependent branch representation with coordinate-dependent trunk functions [3], [4], [15]. Physics-informed learning offers a different inductive bias by penalizing differential-equation residuals [5], [16]. The present comparison intentionally includes both grid operators and pointwise models, since their relative value may change when the output is used for optimization rather than visualization.

Learning-based airfoil optimization has shown large online speedups and has validated selected optima with higher-fidelity solvers [13], [14]. Most such studies develop or test one surrogate family. Our contribution is complementary: data, split, targets, optimizer, starts, bounds, and endpoint solver are held fixed while the model family changes. A matched direct-CFD search then distinguishes prediction error from the effect of optimizing a learned objective.

III. Data and CFD Reference

A. Design space

The dataset contains National Advisory Committee for Aeronautics (NACA) 0006, 0008, 0010, 0012, 0015, 0018, 2412, 2415, 4412, 4415, 4418, and 4421 profiles generated from the standard four-digit parameterization [17]. For each profile, we simulate angles of attack $\alpha \in \{-4, -2, 0, 2, 4, 6, 8, 10\}^\circ$ and Reynolds numbers $Re \in \{10^5, 2 \times 10^5, 5 \times 10^5, 10^6\}$. All cases use liquid water with density 997 kg/m^3 and kinematic viscosity $10^{-6} \text{ m}^2/\text{s}$. With a 1 m chord, the freestream speed is set by $U_\infty = Re\nu/c$; the far-field absolute pressure and vapor pressure are 101325 and 2300 Pa.

Each case is solved with OpenFOAM simpleFoam [18]. For steady incompressible flow, the mean velocity $\bar{u}_i$ and pressure $\bar{p}$ satisfy

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0, \quad (1)$$

$$\rho \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \mu \frac{\partial^2 \bar{u}_i}{\partial x_j \partial x_j} - \rho \frac{\partial \overline{u'_i u'_j}}{\partial x_j}. \quad (2)$$

The Reynolds stress is closed with the k - ω shear-stress transport (SST) model [19]. Angled freestream velocity is imposed at the inlet and far boundary, and OpenFOAM's forceCoeffs supplies total lift and drag after 300 iterations. Cell-centered fields are interpolated to a common 128×64 Cartesian grid over $x/c \in [-1, 2]$ and $y/c \in [-0.75, 0.75]$. Inputs include coordinates, operating parameters, three NACA descriptors, a fluid mask, and signed distance to the profile.

Three simulations with $|C_L|$ or $|C_D| > 5$ are discarded before splitting, leaving 381 cases. All 64 cases from

NACA 0015 and NACA 2412 form the validation set; the remaining 317 cases train the models. Because complete families are reserved, neither validation geometry appears in training at a different operating condition. All retained solves reach iteration 300. At the final iteration, the 95th-percentile maximum initial equation residual is 1.27×10^{-4} ; over the last 50 iterations, the 95th-percentile spans are 0.0059 for C_L and 2.65×10^{-4} for C_D .

B. Targets and cavitation-risk definition

The common output vector contains U_x , U_y , absolute pressure p , eddy viscosity, k , ω , pressure coefficient C_p , cavitation margin, lift coefficient C_L , and drag coefficient C_D . Grid fields use masked normalized mean-squared error, while the case-level force targets use normalized mean-squared error. OpenFOAM's kinematic gauge pressure p_k is converted to absolute pressure as

$$p = p_\infty + \rho p_k, \quad C_p = \frac{p - p_\infty}{\frac{1}{2} \rho U_\infty^2}. \quad (3)$$

For an ambient-pressure value p_a , the pressure-based inception margin is

$$M(x, y; p_a) = p_a + C_p(x, y) \frac{1}{2} \rho U_\infty^2 - p_v. \quad (4)$$

A cell is labeled at risk when $M < 0$. We evaluate this label at $p_a \in \{2500, 3000, 5000, 10000, 25000, 101325\}$ Pa. This sweep creates a useful classification test from the predicted C_p field without representing multiphase physics.

IV. Surrogates and Optimization

A. Model families

The CNN-U-Net has three encoder/decoder levels, skip connections, group normalization, Gaussian error linear unit (GELU) activations, and base width 16. It maps the complete input grid to the output grid [2]. The FNO has width 32, four operator blocks, and 12 retained Fourier modes in each direction. One block is

$$h_{\ell+1} = \sigma(W_\ell h_\ell + \mathcal{F}^{-1}(R_\ell \mathcal{F}(h_\ell))). \quad (5)$$

following the spectral operator construction of [3]. Here, W_ℓ is a local linear map and R_ℓ acts on the retained spectral modes. The DeepONet uses three-layer, width-64 branch and trunk networks with 64 basis functions. For output k ,

$$\hat{y}_k(\mathbf{x}; \boldsymbol{\mu}) = \frac{\mathbf{b}_k(\boldsymbol{\mu})^\top \mathbf{t}_k(\mathbf{x})}{\sqrt{64}} + c_k, \quad (6)$$

as in the branch-trunk construction of [4], where $\boldsymbol{\mu}$ contains the operating and geometry parameters.

The physics-regularized multilayer perceptron (MLP) has three tanh layers of width 64. It is supervised by CFD values at 1,024 sampled points per case. Its extra loss uses the directional residuals

$$r_c = u_x + v_y, \quad r_x = uu_x + p_x/\rho, \quad r_y = vv_y + p_y/\rho, \quad (7)$$

with $\mathcal{L} = \mathcal{L}_{\text{sup}} + 10^{-4} \mathcal{L}_{\text{phys}}$, following the residual-loss principle in [5]. This is a lightweight physics regularizer,

TABLE I

Surrogate representations and measured inference cost per case.

Model	Representation	Principal bias	Params.	ms
CNN-U-Net	grid	local, multiscale	485,178	10.80
Physics-reg. MLP	points	differential residual	9,930	6.88
FNO	grid	global, spectral	595,914	12.07
DeepONet	branch-trunk	operator basis	100,746	13.52

not a data-free physics-informed neural network (PINN) or a complete RANS residual.

Table I summarizes the resulting inductive biases. The two grid models consume the entire Cartesian field at once. DeepONet and the physics-regularized MLP instead evaluate coordinates, which makes their representation naturally pointwise, although the present timing benchmark requests the same complete grid from every model.

All models use AdamW with learning rate 2×10^{-3} , weight decay 10^{-4} , at most 120 epochs, patience 20, gradient clipping, and seed 7. Normalization is computed from training cases only. The best checkpoint is selected by validation loss. Model size ranges from 9,930 to 595,914 parameters (Table II).

Figure 1 reports the complete optimization histories rather than only the selected checkpoints. Because the architectures consume the data in different forms and the objective aggregates heterogeneous normalized targets, loss magnitude should be interpreted within a model, not as a direct cross-model accuracy metric. DeepONet reaches its minimum validation objective at epoch 6, the physics-regularized network at epoch 30, FNO at epoch 54, and CNN-U-Net at epoch 120. The isolated FNO spikes are retained in the record and motivate reporting best-checkpoint results alongside the full history.

Optimization uses dedicated C_L and C_D output heads supervised by OpenFOAM’s total force coefficients. Pressure integration on the coarse common grid is not used for the objective because it omits viscous traction and under-resolves the wall.

B. Why physics regularization rather than a full PINN?

A classical PINN is trained by differentiating its outputs and minimizing the governing-equation and boundary-condition residuals, sometimes with few or no solution labels [5]. Our point network remains primarily supervised by OpenFOAM values; its low-weight residual contains continuity and inviscid directional momentum only. It omits viscous stress, Reynolds stress, the k - ω transport equations, and explicit wall and far-field residuals. Describing it as a full RANS PINN would therefore overstate the physics imposed during training.

This lighter comparator was chosen because it applies a recognizable physical bias while retaining the same labels, split, and modest training budget as the other surrogates. A full hydrofoil RANS PINN is a different experiment: it would require coupled turbulence variables and boundary losses, careful residual balancing near the

TABLE II

Validation performance, model size, and cost. Force RMSE is computed case-wise against OpenFOAM total coefficients.

Model	p RMSE	p R^2	C_L RMSE	C_D RMSE	Params.	Speedup
CNN-U-Net	28.52	0.392	0.0991	0.00453	485,178	$1.148 \times$
Physics-reg. MLP	6.78	0.966	0.0245	0.00204	9,930	$1.801 \times$
FNO	22.73	0.614	0.1383	0.00957	595,914	$1.028 \times$
DeepONet	8.55	0.945	0.0268	0.00184	100,746	$917 \times$

wall, and substantially more automatic-differentiation work. The label “physics-regularized MLP” keeps that distinction visible when interpreting its strong pressure results.

C. Common optimization protocol

At $Re = 5 \times 10^5$, each surrogate drives the same HSO loop. The design vector $\theta = (m, p, t, \alpha)$ contains maximum camber, camber location, thickness, and incidence, with bounds $m \in [0, 0.09]$, $p \in [0.1, 0.9]$, $t \in [0.06, 0.20]$, and $\alpha \in [-4, 10]^\circ$. We launch Nelder–Mead [20] from NACA 0012, 2412, and 4415 and give every run the same 40-iteration budget. The objective is

$$J(\theta) = \frac{C_L}{|C_D| + 10^{-6}} - 10P_M - 5f_M, \quad (8)$$

where P_M penalizes a predicted minimum cavitation margin below 5000 Pa and f_M is the fraction of cells below that margin. The search calls only the surrogate; OpenFOAM is not used inside the loop. Since the nominal-pressure margin stays above the threshold for every selected design, the penalty is inactive in this experiment. We return the best design observed within the fixed budget, select the strongest of the three starts for each architecture, and then generate a new mesh and independent OpenFOAM simulation for that winner.

For the direct-CFD baseline, the same Nelder–Mead procedure starts from the same three foils and uses the same bounds, objective, tolerances, and 40-iteration budget, but every evaluation is obtained by remeshing and running OpenFOAM. This separates endpoint verification, which checks each surrogate’s winner once, from CFD-driven optimization, which supplies a matched reference search.

V. Results

A. Validation accuracy and computational cost

Table II reports results on the 64-case validation set. The physics-regularized network is best for pressure and C_p , with a pressure root-mean-square error (RMSE) of 6.78 Pa and $R^2 = 0.966$. DeepONet gives the best U_x , U_y , C_L , and C_D coefficients. The grid operators are weaker in this low-data setting. FNO retains useful force trends. CNN-U-Net predicts C_p and forces more accurately than absolute pressure or streamwise velocity. The normalized multi-target loss and the affine relationship between p and C_p allow this difference between channels.

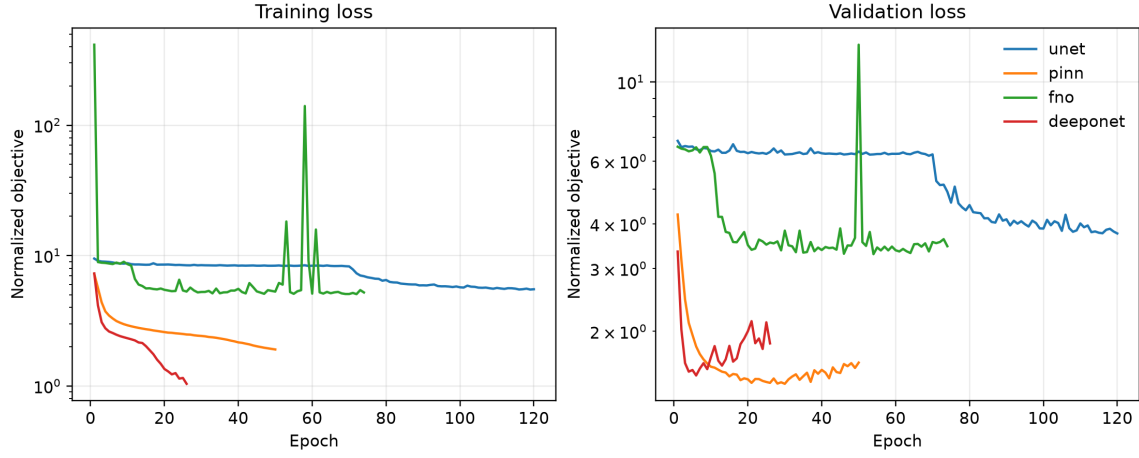


Fig. 1. Training and unseen-geometry validation objectives. Curves terminate after early stopping; objectives combine normalized field and force losses.

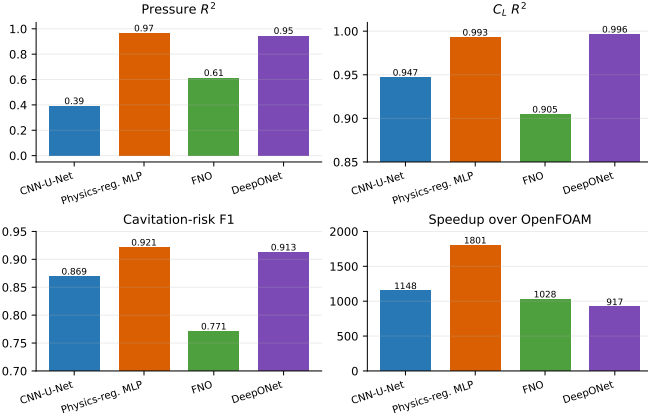


Fig. 2. No architecture dominates every criterion on the 64-case validation set. Cavitation F1 is aggregated over ambient-pressure sweeps.

Measured training times are 868, 104, 547, and 54 s for CNN-U-Net, the physics-regularized network, FNO, and DeepONet, respectively. Median OpenFOAM execution time is 12.4 s per case, while pressure-field inference requires 0.0069–0.0135 s, yielding 917–1801 $\times$ speedups. The benchmark used an 8-core Apple M2 Mac mini with 16 GB memory; PyTorch 2.13 ran on its central processing unit (CPU), and single-process OpenFOAM 9 ran in Docker. These numbers measure repeated local inference and exclude data generation and mesh construction.

The ranking also depends strongly on the target field. Figure 3 shows that the pointwise models recover both velocity components and pressure, whereas CNN-U-Net and FNO are substantially weaker on absolute pressure and U_x . All four models have $R^2 \leq 0.15$ for eddy viscosity, k , and ω . Thus, good C_p and force predictions do not make these checkpoints interchangeable with a complete RANS solution; their demonstrated value is in pressure screening and force-guided design.

TABLE III

Cavitation-risk F1 by ambient pressure. “All” aggregates the six pressure thresholds; no positives occur at or above 5000 Pa.

Model	All	2500 Pa	3000 Pa
CNN-U-Net	0.869	0.874	0.596
Physics-reg. MLP	0.921	0.926	0.383
FNO	0.771	0.774	0.574
DeepONet	0.913	0.920	0.113

Figure 4 shows a representative validation case from the unseen NACA 0015 family. The pointwise models reproduce the leading-edge pressure structure most closely; CNN-U-Net smooths the extrema and FNO introduces spatial texture. The distinct error color scales in the lower row are retained to expose each model’s error structure rather than visually equalizing models with very different ranges.

B. Cavitation-inception screening

The pressure-threshold labels are highly imbalanced, so accuracy is not useful here. The physics-regularized network achieves precision 0.927, recall 0.916, and F1 0.921. DeepONet follows at F1 0.913. CNN-U-Net has the highest recall (0.958) but lower precision (0.796), while FNO obtains F1 0.771. Twenty-four validation case/ambient-pressure combinations contain at-risk cells. This test measures whether a surrogate preserves low- C_p regions well enough for inception screening. It does not model cavity dynamics. That would require multiphase RANS labels such as vapor volume fraction and cavity length [21].

Table III exposes a threshold effect hidden by the aggregate score. Most positive cells occur at 2500 Pa, where the physics-regularized MLP and DeepONet are strongest. At 3000 Pa only 32 true positive cells remain: CNN-U-Net detects 31 but adds 41 false positives, while DeepONet detects only three. Cavitation-screening quality

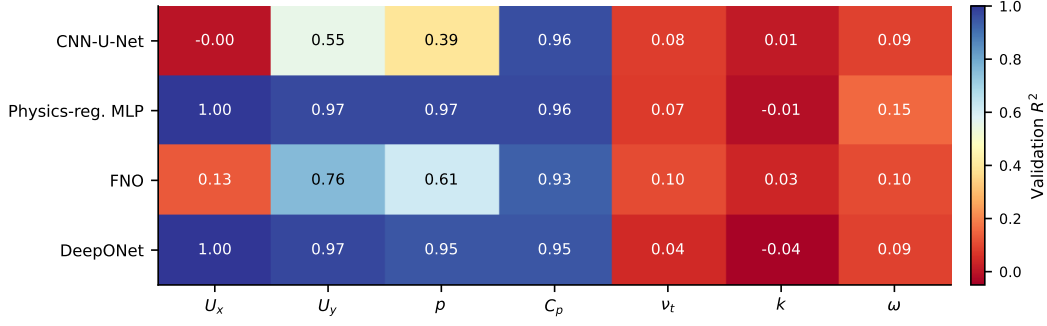


Fig. 3. Full-field coefficient of determination on the unseen-geometry validation set. The turbulence outputs remain difficult for every model even when velocity, pressure, or C_p is accurate.

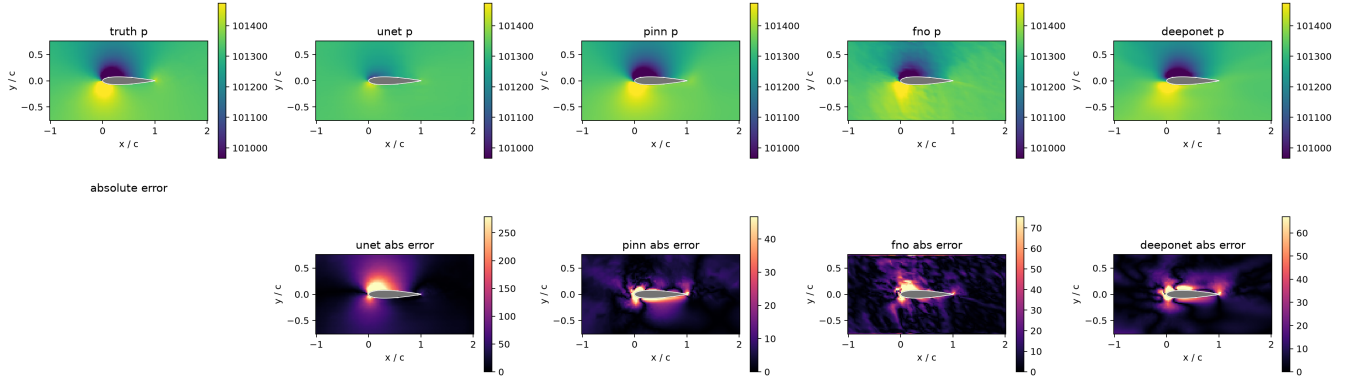


Fig. 4. Absolute-pressure prediction for a representative validation case from the NACA 0015 case (case 160), followed by per-model absolute error. All prediction panels share the truth color scale; error panels use individually labeled scales.

is therefore operating-pressure dependent; the aggregate F1 should not be read as a universal classifier score.

C. Shape optimization against direct CFD

Table V reports the complete L/D comparison: the OpenFOAM value of the starting foil, the surrogate estimate at its selected design, and the result of remeshing that design and solving it once in OpenFOAM. That one-solve check is endpoint validation. Separately, the direct baseline places OpenFOAM inside the matched three-start optimizer, providing a CFD-driven reference under the same local-search budget.

All four surrogate winners improve on their starts. The physics-regularized network moves from $L/D = 25.57$ to 45.96, an absolute gain of 20.39 (79.7%); DeepONet gains 14.97, CNN-U-Net gains 12.55, and FNO gains 0.58. Direct CFD reaches 17.80, 27.33, and 29.89 from NACA 0012, 2412, and 4415, respectively. Its best gain is 4.32 (16.9%). The selected (m, p, t, α) values are $(0.0325, 0.550, 0.060, 3.06^\circ)$, $(0.0667, 0.388, 0.060, 1.24^\circ)$, $(0.0391, 0.372, 0.168, 4.23^\circ)$, and $(0.0335, 0.328, 0.060, 1.60^\circ)$ for CNN-U-Net, the physics-regularized network, FNO, and

DeepONet, respectively. Direct CFD selects $(0.0407, 0.417, 0.144, 3.98^\circ)$.

DeepONet has the smallest winner-level L/D error (2.1%), consistent with its force accuracy, but the physics-regularized network finds the highest-performing shape. CNN-U-Net, the physics-regularized network, and DeepONet all exceed the matched direct-CFD result after endpoint validation; FNO does not. This does not make a surrogate more accurate than CFD. Rather, under a limited local search, the smoother learned objectives lead Nelder–Mead into different basins.

VI. Discussion

DeepONet’s 2.1% endpoint error confirms that force calibration carries into HSO, but it does not find the best design: the physics-regularized network reaches $L/D = 45.96$ rather than 40.86. CNN-U-Net and FNO both underpredict their selected endpoints by about 20%, yet only CNN-U-Net locates a substantial gain. Candidate ranking and value calibration are therefore distinct model properties.

The direct-CFD history sharpens that distinction. It spends 266 evaluations and 38.5 minutes of summed wall time, yet its best result is $L/D = 29.89$. Endpoint-

TABLE IV

Best L/D objective found from each starting foil. Surrogate rows report learned objective values inside their search; direct CFD reports OpenFOAM values. Bold entries are the starts selected for endpoint validation.

Evaluator	NACA 0012 start	NACA 2412 start	NACA 4415 start
CNN-U-Net	17.82	30.17	28.08
Physics-reg. MLP	22.35	43.55	43.62
FNO	16.18	18.44	21.11
DeepONet	33.63	39.99	36.17
Direct CFD	17.80	27.33	29.89

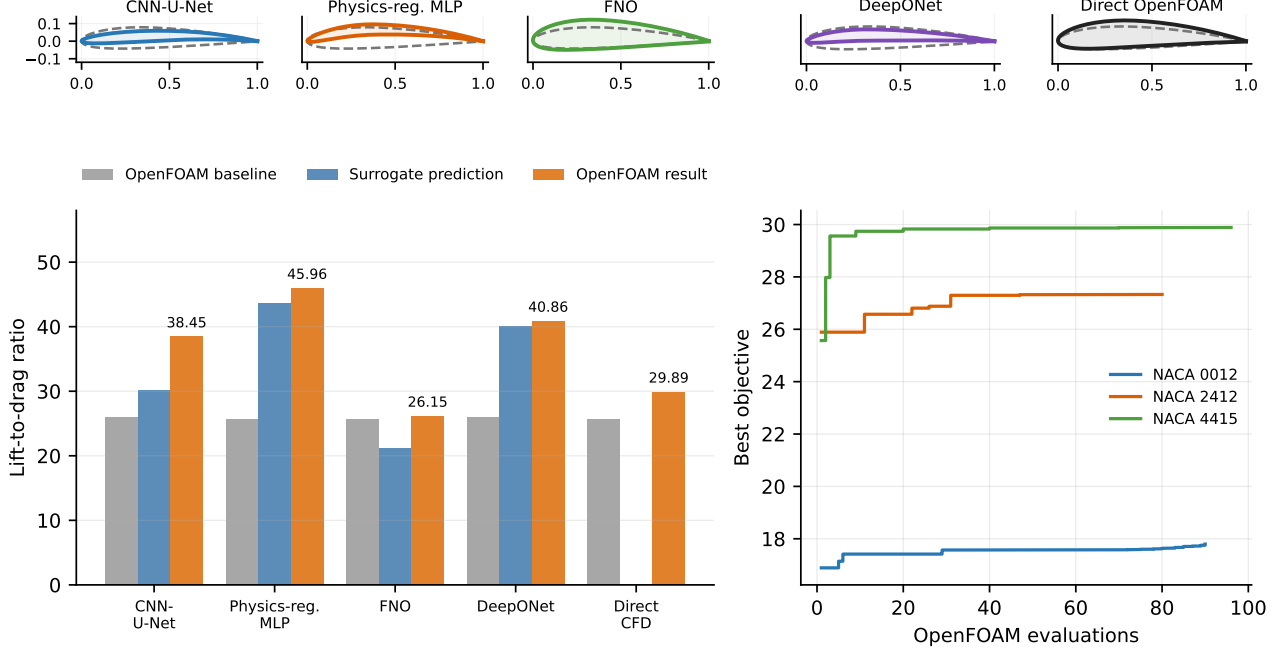


Fig. 5. Top: selected shapes against the same dashed NACA 2412 geometric reference. The common reference supports shape comparison and is not necessarily the optimizer initializer; actual initializers are listed in Table IV. Bottom left: surrogate predictions, CFD endpoint checks, and the direct-CFD result. Bottom right: best-so-far objective during each direct OpenFOAM search.

TABLE V

HSO results at $Re = 5 \times 10^5$. The baseline and final columns are independent OpenFOAM results; Δ is their absolute difference.

Model	Start	Base	Pred.	Final	Δ	Gain
CNN-U-Net	2412	25.89	30.17	38.45	+12.55	48.5%
Physics-reg. MLP	4415	25.57	43.62	45.96	+20.39	79.7%
FNO	4415	25.57	21.11	26.15	+0.58	2.3%
DeepONet	2412	25.89	39.99	40.86	+14.97	57.8%
Direct CFD	4415	25.57	-	29.89	+4.32	16.9%

validated CNN-U-Net, DeepONet, and physics-regularized designs exceed that value by 8.56, 10.97, and 16.07, respectively. The direct run is a matched local-search baseline, not a claim of the global CFD optimum; its value is that it isolates how the evaluator changes search behavior at fixed settings.

The three starts also expose different objective landscapes. The physics-regularized model predicts nearly identical optima from NACA 2412 and 4415 ($L/D = 43.55$

and 43.62) but only 22.35 from NACA 0012; FNO remains between 16.18 and 21.11 across all starts. The three stronger endpoint designs approach 6% thickness, whereas FNO selects 16.8%. All winners retain about 99 kPa of nominal predicted cavitation margin, so the penalty is inactive: the present HSO result is an L/D experiment accompanied by cavitation screening, not a demonstration that cavitation changes the optimum.

Table IV also shows why reporting only one optimization start would be misleading. Three models and direct CFD favor a cambered start, but the preferred family differs, and the spread across starts ranges from 4.94 for FNO to 21.27 for the physics-regularized MLP. Because these are predictions on different learned objectives, their magnitudes are not a cross-model accuracy ranking. They quantify search sensitivity within each evaluator; the fresh CFD endpoint values in Table V remain the common comparison.

The field results help explain why parameter count alone is not predictive. The 9,930-parameter physics-regularized

MLP is the smallest model and gives the best pressure result, while the 595,914-parameter FNO is the largest and ranks third on pressure. In this dataset, an operator model receives only 317 complete training realizations even though each realization contains many grid cells. The pointwise networks turn those same cases into supervised coordinate samples and encode the operating parameters directly. This is favorable for a fixed grid and a low number of geometries, but it does not imply that pointwise models will dominate as geometric diversity, resolution, or three-dimensional domain size grows. The comparison instead identifies the inductive bias that works best at the present data scale.

Runtime should likewise be interpreted at the workflow level. Repeating the 266 evaluations used by the matched direct search would require about 38.5 minutes of CFD wall time, compared with approximately 1.8–3.6 seconds of surrogate inference. Even including training, the measured online cost breaks even after roughly 5 DeepONet, 9 physics-regularized MLP, 45 FNO, or 71 CNN-U-Net evaluations. Those counts exclude CFD data generation, which is the dominant up-front expense and must be amortized across operating sweeps or multiple design studies. The speedups therefore support reuse of a trained surrogate, not replacement of the simulations needed to construct it.

The water operating conditions also clarify the cavitation result. At $Re = 5 \times 10^5$ and the stated 1 m chord, $U_\infty = 0.5$ m/s and the dynamic pressure is only about 125 Pa. With nominal ambient pressure near 101 kPa, none of the optimized designs approaches vapor pressure, so the cavitation penalty cannot influence the optimum. The lower-pressure sweep tests whether the networks preserve the leading-edge low-pressure pattern under an inception criterion; it should not be interpreted as a prediction of a developed vapor cavity. A future cavitation-constrained design study would choose operating pressure and velocity jointly so that inception is active within the search space.

Finally, three surrogate winners approach the lower thickness bound. This is a common place for learned-objective error to matter because the optimizer is querying continuous shapes that were not simulated during training. The fresh mesh-and-solve endpoint check answers the immediate question—whether the selected shape still performs in the reference solver—without claiming that the learned surface is uniformly accurate around that point. Local CFD samples around each winner, structural thickness constraints, or a trust-region update would be natural additions when the objective is a deployable design rather than the controlled benchmark considered here.

The comparison uses one training seed, two-dimensional steady RANS, and the NACA four-digit space. A formal grid-convergence study, external force validation, and repeated training seeds remain the highest-value checks.

VII. Conclusion

We followed four neural surrogates through the complete path from RANS-field prediction to a design decision. On unseen NACA families, the physics-regularized point network gives the best pressure result ($R^2 = 0.966$) and cavitation-inception F1 score (0.921), while DeepONet gives the most accurate lift and drag. Every model evaluates a pressure field in 6.9–13.5 ms, corresponding to 917–1801 $\times$ speedup over the measured OpenFOAM runs.

The HSO experiment changes the ranking from a simple accuracy table. DeepONet predicts its winner most accurately, but the physics-regularized network finds the best design, increasing OpenFOAM L/D from 25.57 to 45.96. CNN-U-Net and DeepONet also find substantial gains, from 25.89 to 38.45 and 40.86, respectively, whereas FNO reaches only 26.15 from 25.57. The matched direct-CFD search reaches 29.89, showing that a fast surrogate can change not only search cost but also which basin a budgeted local optimizer reaches. A useful scientific surrogate should therefore be judged on both its validation predictions and the consequence of acting on them. In this benchmark, the strongest model is not merely the one with the lowest aggregate loss; it is the one that combines accurate pressure screening, fast evaluation, and reliable guidance inside the design loop.

Artifact Availability

Code, configurations, split definitions, per-case metrics, optimization traces, and exact execution commands are public at <https://github.com/aaravvd/hydrofoil-neural-surrogates>. The CFD dataset and trained checkpoints are archived at <https://doi.org/10.5281/zenodo.21845241>. The repository records the software environment and checksums needed to connect the source release to the archived artifacts.

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OpenAI Codex was used throughout manuscript preparation as a general-purpose writing and code-assistance tool, including language editing, plotting-script development, and reproducibility checks [22]. The author reviewed the manuscript, reran the reported analyses, and accepts full responsibility for the scientific claims and submitted text.

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